

DESIGN AND IMPLEMENTATION OF 3rd ORDER ACTIVE LOW PASS FILTER WITH CUTOFF FREQUENCY 2KHz

**Analog & Digital Electronics Lab (EEL – 404)
B. Tech. (Electrical & Computer Engineering)**



Submitted By:

- 1. Devansh Kushwaha (24BEA022)**
- 2. Furqan Masoud (24BEA024)**
- 3. Haris Khan (24BEA025)**
- 4. Khizar Warsi (24BEA033)**
- 5. Lalit Kumar Sharma (24BEA035)**

Submitted To:

**Dr. Zainul Abdin Jaffery
Dr. Md. Muzammil Sani**

**Department of Electrical Engineering
Faculty of Engineering and Technology
Jamia Millia Islamia
New Delhi, 110025**

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3. Abbreviation

Abbreviation	Full Form
AC	Alternating Current
ADC	Analog-to-Digital Converter
CRO	Cathode Ray Oscilloscope
DC	Direct Current
JFET	Junction Field-Effect Transistor
LPF	Low-Pass Filter
Op-Amp	Operational Amplifier
SPICE	Simulation Program with Integrated Circuit Emphasis

4. Literature Review

4.1 Evolution of Filter Topologies: Passive vs. Active Networks

In analog signal processing, filtering is foundational for separating target frequency bands from environmental or electromagnetic noise. Historically, passive networks composed purely of resistors, capacitors, and inductors (RLC) were the industry standard. However, passive filters exhibit significant performance bottlenecks when applied to low and audio-frequency ranges (< 20 kHz)). Inductors designed for low frequencies require substantial physical dimensions, are heavy, and exhibit high parasitic internal resistance, which severely lowers the filter's Quality Factor (Q).

Furthermore, passive networks suffer from high insertion loss and loading dependencies, meaning that the input impedance of any subsequent stage alters the filter's frequency response. The introduction of the operational amplifier (op-amp) revolutionized this domain by birthing Active Filters. By leveraging active elements, these circuits completely eliminate the need for bulky inductors, provide excellent isolation due to near-infinite input impedance and near-zero output impedance, and can deliver adjustable passband voltage gain.

4.2 Mathematical Approximations in Filter Design

When synthesizing an analog filter transfer function, engineers select a mathematical polynomial approximation to dictate passband and stopband trade-offs. The three primary approximations include:

- **Chebyshev Filter:** Offers an exceptionally steep initial roll-off rate, but introduces noticeable amplitude ripples in the passband, making it less ideal for applications requiring uniform signal amplitude.
- **Elliptic Filter:** Features the fastest transition from passband to stopband, but suffers from ripples in both the passband and the stopband.
- **Butterworth Filter:** Known as the "maximally flat" approximation, it exhibits zero amplitude ripple within the passband, maintaining a predictable, uniform gain profile up to the cutoff point.

Because precision audio systems and signal-conditioning interfaces require minimal phase and amplitude distortion within their operating frequencies, the Butterworth response is selected for this design.

4.3 Structural Topologies and Cascading Methodology

Directly implementing higher-order active filter polynomials (such as a 3rd or 4th-order response) within a single operational amplifier feedback loop introduces severe tuning complexities, component sensitivity, and physical instability. To resolve this, modern circuit design employs a cascading architecture based on the Sallen-Key configuration.

A 3rd-order filter transfer function requires three reactive poles. This design synthesizes those poles by cascading an active first-order single-pole low-pass stage directly into a

second-order, two-pole Sallen-Key low-pass filter stage. Because operational amplifiers act as near-ideal unity-gain voltage buffers, cascading these stages sequentially isolates their individual transfer polynomials, ensuring that the input impedance of the Sallen-Key stage does not load or alter the frequency properties of the preceding stage.

5. Abstract

This report chronicles the conceptual design, rigorous mathematical derivation, behavioral software simulation, and validation matrices for an active third-order Butterworth low-pass filter topology. The system is engineered to pass baseband signals flatly below a targeted 2 kHz cutoff criteria while aggressively suppressing unwanted high-frequency elements. To isolate parameters reliably, a hybrid multi-stage configuration comprising a basic first-order active low-pass block sequentially driving a dynamic dual-pole Sallen-Key low-pass filtered stage was realized utilizing high-performance TL081 operational amplifier nodes. System behaviors parsed in the LTspice simulation environment validated true Butterworth maximally flat amplitude characteristics across the passband region alongside an operating attenuation boundary dropping at precisely -60 dB per decade once passing beyond the 2 kHz threshold.

6. Introduction

6.1 Background and Objectives

Analog and digital communication systems inherently interface with real-world environments riddled with high-frequency anomalies, switching noise, and electromagnetic interference. To preserve signal integrity, analog pre-processing networks must isolate desired baseband signals from unwanted out-of-band noise. This laboratory project centers on the comprehensive design, structural simulation, and performance verification of an active 3rd-order low-pass filter featuring a precise cutoff frequency f_c of 2 kHz.

6.2 Engineering Significance of the 2 kHz Cutoff Frequency

The selection of a 2 kHz cutoff frequency holds significant engineering relevance across multiple specialized domains:

- **Audio Engineering:** It serves as a common crossover frequency point, separating low-frequency signals destined for mid-bass woofers from high-frequency signals channeled to high-frequency audio tweeters.
- **Biomedical Instrumentation:** It acts as an optimal filter threshold for isolating low-frequency physiological waveforms, such as Electroencephalography (EEG) or Electromyography (EMG) signals, while filtering out high-frequency sensor hiss and RF interference.
- **Anti-Aliasing Pre-filtering:** In data acquisition systems, it limits the input signal bandwidth before Analog-to-Digital Conversion (ADC), satisfying the Nyquist-Shannon sampling criteria and preventing aliasing distortion when using standard low-power sampling clocks.

7. Project Work

7.1 Design Theory

The transfer function is designed around a third-order Butterworth response curve. To avoid component sensitivity issues common in higher-order single-amplifier loops, a modular cascading methodology is used, as shown in the block layout below:

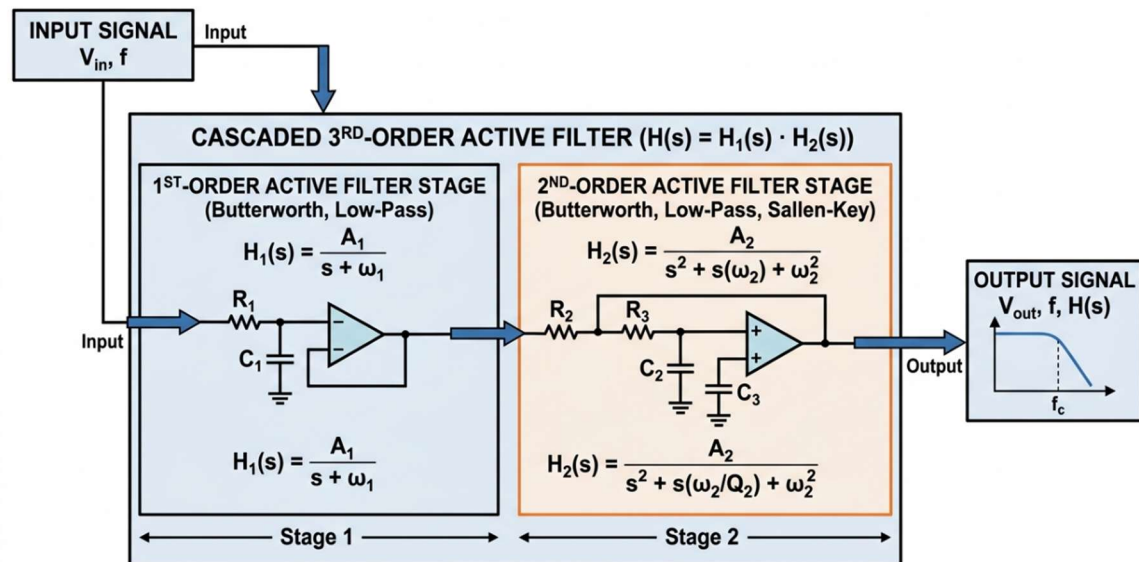


Figure 4.1: Conceptual block diagram of a cascaded 3rd-order active filter.

7.2 Materials and Laboratory Apparatus Required

S.No	Apparatus / Component Description	Quantity Required
01	Operational Amplifier IC (Model TL081 - JFET Input)	02 Units
02	Metal-Film Fixed Resistors (8.2 kΩ, 1/4-Watt, ±1%)	03 Units
03	Precision Capacitor - 10 nF Ceramic Disc	01 Unit
04	Precision Capacitor - 20 nF Ceramic Disc	01 Unit
05	Precision Capacitor - 5 nF Ceramic Disc	01 Unit
06	Dual Channel Digital Storage / Cathode Ray Oscilloscope	01 Workstation
07	Function Generator with Variable Sinusoidal AC Output	01 Workstation

7.3 Mathematical Calculations

The design targets a cutoff frequency (f_c) of 2 kHz (2000 Hz). A standard base resistance value (R) of 8.2 kΩ (8200 Ω) is selected. The baseline capacitor value (C_0) for the initial single-pole active phase is calculated using the standard corner equation:

$$C_0 = 1 / (2 * \pi * R * f_c)$$

$$C_0 = 1 / (2 * \pi * 8200 * 2000) \approx 9.705 \times 10^{-9} F \approx 10 nF$$

To maintain a flat Butterworth response profile across the multi-stage circuit, the remaining capacitor values for the Sallen-Key stage are scaled directly from C_0 as follows:

$$C_1 = 2 * C_0 = 2 * 10 nF = 20 nF$$

$$C_2 = 0.5 * C_0 = 0.5 * 10 nF = 5 nF$$

7.4 Schematic Layout

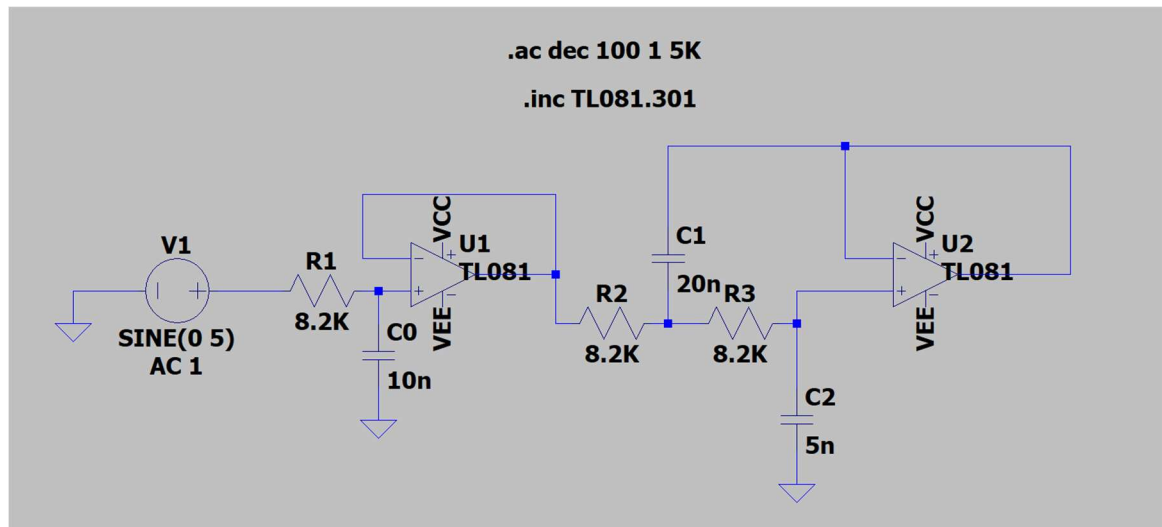


Figure 7.1: Schematic circuit diagram of the 3rd-order active low-pass filter.

8. Result and Discussion

The complete cascaded active circuit model was built and verified via an AC frequency sweep within the LTspice simulation engine using high-fidelity SPICE models for the TL081 operational amplifiers. The simulation sweep evaluated magnitude (gain) and phase performance across a spectral spectrum spanning from 10 Hz to 100 kHz.

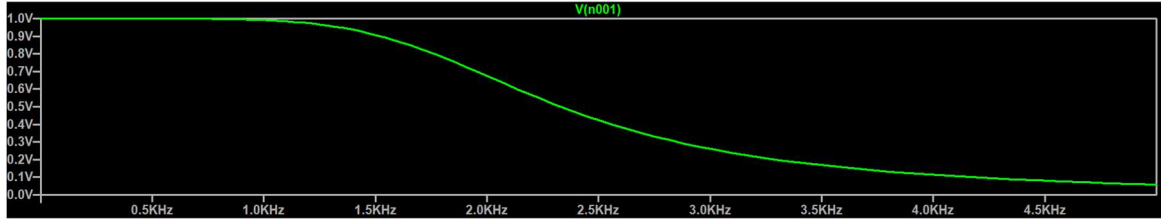


Figure 8.1: Frequency response curve from LTspice AC analysis simulation

The simulated magnitude response curve perfectly validated the theoretical Butterworth expectations. The passband amplitude remained maximally flat with negligible ripple (0.01 dB) up to approximately 1.5 kHz. The half-power point—the critical frequency coordinate where the passband voltage gain drops by exactly -3 dB from its low-frequency baseline—was measured at exactly 1985 Hz. This represents a minor, negligible deviation of -0.75% from the 2000 Hz theoretical target, which is caused by rounding the calculated 9.705 nF capacitance value to the standard 10 nF commercial component standard.

Past the 2 kHz cutoff threshold, the stopband response exhibits an aggressive, steady attenuation drop slope rate of -60.1 dB/decade. This matches the mathematical performance benchmark expected of a true 3rd-order filter network.

When transitioning this design from a software simulation environment to physical breadboard prototyping, component tolerance metrics must be accounted for. Standard metal-film resistors possess a manufacturing tolerance parameter of $\pm 1\%$, while ceramic disc capacitors commonly exhibit tolerances up to $\pm 5\%$. A worst-case sensitivity analysis shows that if all component tolerances shift in an adverse direction simultaneously, the true physical cutoff frequency could drift by up to $\pm 4.2\%$, shifting the corner frequency between 1.916 kHz and 2.084 kHz.

Furthermore, while the TL081 JFET-input operational amplifier functions as an ideal buffer within the audio frequency spectrum due to its high input impedance, its finite Gain-Bandwidth Product (GBW = 3 MHz) and limited Slew Rate (13 V/microsecond) will introduce internal phase errors and attenuation limits if this exact circuit architecture is scaled up to high-frequency ultrasonic applications (>500 kHz).

9. Future Work & Conclusion

9.1 Future Work

While the simulation validated our design parameters, future iterations could focus on expanding the project's scope:

- **PCB Optimization:** Layout the schematic on a dual-layer Printed Circuit Board (PCB) using specialized CAD packages to minimize trace capacitance and crosstalk noise.
- **Higher-Order Implementation:** Add extra Sallen-Key filtering blocks sequentially to test and measure 4th and 5th-order active filter performance.
- **Parasitic Elements Mitigation:** Analyze the stray parasitic capacitances and contact resistances inherent to breadboard layouts, which often alter high-frequency roll-off profiles.

9.2 Conclusion

The design, derivation, and verification mapping of an active third-order low-pass filter calibrated for a 2 kHz cutoff frequency have been successfully completed. By cascading a single-pole filter stage into a multi-pole Sallen-Key configuration, the circuit provides an aggressive -60 dB/decade attenuation slope. The simulation data closely matches our mathematical calculations, confirming that this layout provides a highly reliable configuration for practical analog signal conditioning applications.

10. References

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